STAT 236 Project Proposal

From Purchases to Communities
How User Activity Affects Product Network Construction

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Project type: Implementation and comparison of existing methods through theoretical calculations and simulations with known ground truth.

Timeline 1 submission: November 8, 2026

Problem and motivation

User–product records form a bipartite network. Products can be connected by shared purchasers, and community detection can identify groups with similar purchasing patterns. However, a user purchasing d distinct products contributes to $d(d-1)/2$ product pairs. A small number of active users may therefore dominate a projected product network.

High activity need not be harmful. A frequent buyer with focused preferences may provide useful category information, whereas one who buys broadly may obscure it. We study when activity weighting improves community recovery and when it discards useful information. The groups represent purchasing preferences, not necessarily functional similarity between products.

Our main questions are:

(1) How do activity heterogeneity and preference breadth interact with projection weights?

(2) When does activity weighting help relative to counts and Jaccard similarity?

(3) How do these methods compare with a normalized SVD baseline operating directly on the bipartite matrix?

Background and related methods

Newman (2001) weights a contribution to a collaboration network by the inverse of the number of other participants, motivating our $1/(d-1)$ projection weight. Zhou et al. (2007) study weighted bipartite projection through resource allocation; their method provides context rather than being identical to our symmetric weighting. Dhillon (2001) develops bipartite spectral partitioning, motivating a direct normalized SVD baseline. Rohe et al. (2016) provide related spectral co-clustering ideas for asymmetric networks. We use these existing methods to investigate a focused statistical question, rather than proposing a new algorithm.

Methods and experiments

Data generation and initial parameters

Let B be an $n \times p$ binary matrix, with $B(u, i) = 1$ when user u purchases product i. Initially, $n = 1{,}000$, $p = 300$, and $K = 3$ equal product categories. Users have preferred categories assigned independently of activity and approximately equally represented. All products within a category have equal sampling weight.

Each user purchases d(u) distinct products by weighted sampling without replacement. A product in the preferred category has weight r(u); other products have weight 1. Baseline preference strength is $r = 2$, with pilot values 1.25, 2, and 4. In the focused condition, $r(u) = r$ for everyone. In the broad-active condition, the most active 20% have $r(u) = 1$, and other users retain r. Ties at the activity cutoff are broken randomly, independently of category. Activity and preference settings are crossed so activity alone is not assumed to reduce information.

Mean purchases	Low heterogeneity	Medium heterogeneity	High heterogeneity
6	All users: 6	80%: 3; 20%: 18	80%: 1; 20%: 26
12	All users: 12	80%: 6; 20%: 36	80%: 2; 20%: 52

These initial purchase-count distributions have exactly matched means within each density setting. Counts are assigned independently of preferred category. In the low-heterogeneity setting, the active 20% in the broad-active condition are a random subset, giving a control for preference breadth without degree variation. Realized purchase counts, category balance, and isolated products will be checked. Final parameters will be frozen after the pilot.

Three product network constructions

For distinct products i and j, define shared(i, j) = $\Sigma_u B(u, i)B(u, j)$ and buyers(i) = $\Sigma_u B(u, i)$. We compare:

$$W_{count}(i, j) = \Sigma_u B(u, i) B(u, j)$$

$$W_J(i, j) = \text{shared}(i, j) / [\text{buyers}(i) + \text{buyers}(j) - \text{shared}(i, j)]$$

$$W_{weighted}(i, j) = \Sigma_{u: d(u) \geq 2} B(u, i) B(u, j) / [d(u) - 1]$$

Jaccard is zero when its denominator is zero. Users with fewer than two purchases contribute no pairs. All projected diagonals are zero; weighted edges are retained without thresholding. For each projection, we use the same normalized spectral procedure: form $D^{(-1/2)} W D^{(-1/2)}$, extract the K largest algebraic eigenvectors, normalize nonzero rows, and run k-means with 20 initializations and recorded seeds. Zero-strength products or zero embedding rows are unassigned and count as primary-metric errors.

Direct bipartite baseline

As a fourth method, form $M = D_u^{(-1/2)} B D_p^{(-1/2)}$, where D_u and D_p contain user and product degrees. Remove zero-degree rows and columns, take the leading K right singular vectors, normalize their nonzero rows, and apply the same k-means settings to cluster products. We call this a normalized bipartite SVD baseline inspired by Dhillon (2001), rather than claiming an exact implementation of the original recursive co-clustering algorithm. K is fixed at its true value for every method.

An unnormalized SVD of B and eigen-decomposition of B^TB share right-eigenvector geometry. Moreover, M^TM is itself a normalized weighted Gram matrix. Thus this comparison examines normalization and representation choices; it does not prove that direct SVD retains information unavailable to every possible projection. We will document diagonal treatment because projected methods remove self-weights while the direct baseline retains their implicit effect.

Theoretical calculations and predictions

We will derive expected within-category and between-category weights using the exchangeability of products inside each category. Let s = p/K and C(u, k) be the number of products user u purchases from category k. For distinct products, define the conditional pair-inclusion probability q(u, i, j), conditional on activity and preferred category. Then:

Same category k: $q(u, i, j) = E[C(u, k)(C(u, k) - 1)] / [s(s - 1)]$.

Different categories k and l: $q(u, i, j) = E[C(u, k)C(u, l)] / s^2$.

$E[W_{count}(i, j)] = \Sigma_u q(u, i, j)$.

$E[W_{weighted}(i, j)] = \Sigma_{u: d(u) \geq 2} q(u, i, j) / [d(u) - 1]$.

The expectations above are conditional on user parameters. The category-count moments reflect the actual sampling-without-replacement scheme; we will derive tractable special cases and estimate remaining moments by simulation. In particular, a completely unfocused user samples a uniform

subset, giving $q(u, i, j) = d(u)(d(u) - 1)/[p(p - 1)]$ for any pair. This user adds no expected within-versus-between contrast.

Exactly, a user contributes total unordered-pair weight $d(d - 1)/2$ under Count and $d/2$ under activity weighting when $d \geq 2$; a user with $d < 2$ contributes zero to both. This establishes quadratic versus linear growth in total influence. It is not a universal statement about each pair or a proof of clustering failure after spectral normalization.

For Jaccard, the expected ratio is not generally the ratio of expectations. We will use the latter only as an explicitly labeled approximation, checked against simulated mean weights. We predict that broad-active users can add a disproportionate background under Count, while weighting may reduce this effect. Focused active users can instead strengthen the category signal. We will compare expected block contrast and empirical variability with recovery errors rather than claim a universal threshold.

Main design and evaluation

The main grid crosses three activity distributions, two preference settings, and two mean purchase counts: 12 cells. We plan 100 independent datasets per cell. All four methods use the same datasets, giving paired comparisons.

The primary metric is misclustering rate after optimal label matching, including unassigned products as errors. Secondary metrics are adjusted Rand index among assigned products, assignment coverage, within-minus-between mean edge weight, product strength variation, and runtime. We report Monte Carlo uncertainty and paired method differences using datasets as the independent unit. Reported intervals will describe variability across generated datasets.

Jaccard and Count may be nearly equivalent when product degrees are similar and shared counts are small relative to buyer counts. We will quantify their edge-weight rank association and clustering agreement. A small method difference is a result, not a reason to remove cells. Unequal product popularity is an optional sensitivity analysis; it will be labeled separately rather than introduced solely to force Jaccard to differ.

Pilot checkpoint and scope control

By October 25, run 10 datasets per initial cell, verify purchase totals and weighting identities, check label matching and zero-degree handling, and record runtimes. Use the same datasets for every method. The pilot is for implementation and difficulty calibration, not for claiming a winning method; formal simulations use fresh seeds.

The initial design is accepted if the pipeline runs reliably, paired comparisons are reproducible, and at least two cells have a median misclustering rate between 0.05 and 0.40 for at least one method. This difficulty criterion does not require a performance gap. Cells that are trivial or near chance remain in the study as context. Coverage and runtime must be reported, and the full 1,200-dataset grid should fit an estimated 24-hour compute budget.

If every cell is trivial or near chance, use a bounded calibration grid with mean counts 3, 6, 12, and 24 and r values 1.25, 2, and 4, preserving matched means across heterogeneity levels. Select an informative difficulty range without choosing for the largest method gap, then freeze settings on October 26. If no useful range emerges, revisit the generator before full runs. Direct SVD or weighting need not underperform or win for the project to proceed.

Limitations

Projected edge weights are dependent because one user contributes to multiple product pairs. We do not assume an independent-edge SBM for the projection. Balanced categories and equal within-category popularity simplify the main simulation, limiting generalization. Interpretation will distinguish raw expected edge contrast, spectral normalization, sampling noise, and clustering performance.

Relation to the course and planned outputs

The project applies bipartite networks, degree heterogeneity, eigen-decomposition, SVD, spectral clustering, and misclustering error. It fits the explicitly permitted format of implementation and comparison of existing methods. The group will confirm that it does not substantially overlap with members' existing research.

The report will prioritize four displays: (1) a schematic of users, products, and projection weights; (2) theoretical and simulated user influence and block contrast; (3) recovery-error curves across heterogeneity with all four methods and uncertainty intervals; and (4) paired method differences, including Count–Jaccard agreement. The webpage will summarize the question, show one network example and the main recovery figure, explain the principal finding, and link reproducible code. It will not require a live recommender system.

Timeline and division of labor

Dates	Milestone
Oct. 18–25	Literature, generator, four methods, theory, pilot checkpoint
Oct. 26–30	Freeze parameters; run main simulations with fresh seeds
Oct. 31–Nov. 3	Interpret results and figures
Nov. 4–7	Write and revise report; build concise webpage
Nov. 8	Submit report and webpage

Four roles will be assigned within the group: generator and theoretical calculations; projection methods and direct SVD; evaluation and figures; literature and webpage. All members write and review the report. Supporting materials include code, package versions, random seeds, pilot settings, final parameter choices, and instructions to reproduce figures.

References

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